

SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Final Project Report

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Program: Infosys Virtual Internship

Project: SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

Framework: YOLOv8 (Ultralytics), Streamlit, OpenCV

Hardware Used: NVIDIA GTX 1650 Ti GPU (4GB)

Software: Python, PyTorch, Ultralytics YOLOv8, OpenCV, Streamlit, VS Code

Abstract

SafetyEye is an AI-powered workplace safety monitoring system designed to detect safety compliance in real-time using computer vision. The system identifies whether workers are wearing essential safety equipment such as helmets, masks, and safety vests. It also detects objects like machinery and vehicles to enhance situational awareness.

The project evolved from basic dataset preparation and model training to a fully integrated real-time monitoring dashboard with analytics and logging features. This system demonstrates how artificial intelligence can be applied in industrial environments to reduce risks and improve safety standards.

Project Objectives

- Detect safety equipment using deep learning
 - Identify safety violations in real-time
 - Build a real-time monitoring system using webcam and video input
 - Develop an interactive dashboard for monitoring and analysis
 - Maintain logs of safety violations
 - Provide analytics and insights for safety improvement
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Milestone 1 – Dataset Preparation and Initial Training

Overview

Milestone 1 focused on setting up the development environment, exploring the dataset, and performing initial model training.

Dataset Details

- **Total Images:** 2801
- **Classes:** 10

Classes

- Hardhat
 - Mask
 - NO-Hardhat
 - NO-Mask
 - No-Safety Vest
 - Person
 - Safety Cone
 - Safety Vest
 - Machinery
 - Vehicle
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Dataset Split

Split	Images
Training	1960
Validation	560
Testing	281

Initial Training Configuration

Parameter Value

Model YOLOv8n

Epochs 1

Image Size 640

Initial Results

Metric Value

Precision 0.464

Recall 0.330

mAP@0.5 0.314

mAP@0.5:0.95 0.170

Outcome

- Dataset verified and cleaned
 - Training pipeline successfully tested
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Milestone 2 – Model Training and Optimization

Overview

Focused on improving model performance through training and experimentation.

Best Model Configuration

Parameter Value

Model YOLOv8s

Epochs 50

Image Size 640

Best Results

Metric Value

Precision 0.885

Recall 0.730

mAP@0.5 0.802

Experiments Conducted

- Increased epochs (100)
 - Increased image size (800)
 - Larger model (YOLOv8s)
 - Data augmentation
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Outcome

- Improved detection accuracy
 - Selected best-performing model
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Milestone 3 – Real-Time Detection System

Overview

Converted trained model into a real-time system using OpenCV.

Features Implemented

- Webcam-based detection
 - Video-based testing
 - Person-wise safety analysis
 - Real-time alert generation
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Safety Rules

- Helmet Missing
 - Mask Missing
 - Safety Vest Missing
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Results

- Real-time detection achieved
 - Alerts generated accurately
 - Smooth performance on live video
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Outcome

- Fully functional real-time safety monitoring system
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Milestone 4 – Dashboard, Analytics and System Integration

Overview

Final transformation into a complete application with UI, analytics, and logs.

Dashboard Features

- Streamlit-based UI
 - Sidebar navigation
 - Real-time video feed
 - KPI metrics (persons, unsafe, alerts)
 - Start/Stop monitoring
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Analytics Features

- Violations over time
- Violation distribution
- Safety compliance rate

- AI insights

Logs System

- Stores all events
- Filtering and search
- CSV export
- Color-coded records

System Architecture

- Input → Webcam/Video
- Processing → YOLOv8
- Logic → Safety detection
- Storage → Session logs
- UI → Streamlit dashboard

Performance Summary

Feature	Status	Performance Metric
GPU Inference	Complete	~25–30 FPS
CPU Inference	Complete	~5–8 FPS
UI Response	Complete	<0.5 sec
Log Accuracy	Complete	100%

Outcome

- Full application developed
 - Smooth and interactive system
 - Real-time analytics enabled
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System Evolution

Stage	Description	Level
M1	Dataset + Initial Training	Basic
M2	Model Optimization	Intermediate
M3	Real-Time Detection	Advanced
M4	Dashboard + Analytics + Logs Production	

Challenges Faced

- Incorrect dataset labels
- Detection misclassification
- Real-time performance lag
- UI integration complexity

Solutions Implemented

- Fixed dataset labeling issues
- Improved detection logic
- Optimized processing pipeline
- Modularized code structure

Future Improvements

- Multi-camera support
 - Cloud deployment
 - Database integration
 - Alert notifications (SMS/Email)
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Conclusion

The SafetyEye project successfully demonstrates the application of artificial intelligence in improving workplace safety. Starting from dataset preparation to building a complete real-time monitoring system, the project achieved all its objectives.

The final system integrates detection, alert generation, analytics, and logging into a single platform, making it suitable for real-world deployment. This project highlights the potential of computer vision systems in reducing workplace hazards and enhancing safety compliance.